

CS224 Object Oriented Programming and Design Methodologies Lab Manual



QUIZ 2 - POINTERS AND CLASSES

DEPARTMENT OF COMPUTER SCIENCE

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Quiz guidelines

You will attempt this quiz on your lab machines. Your submission should be a single cpp file named *firstname.lastname_quiz2.cpp*. The file should contain all of code and implementation of of the quiz problems. You may use the following documentation to look up syntax <https://en.cppreference.com/w/>. Accessing any site other than hulms and the provided documentation is prohibited. You are not allowed to communicate with your peers or use any external resources. You can not refer to your notes, books or lecture slides during the quiz. Use of generative AI is strictly prohibited. Not abiding by any of the rules will result in an immediate zero in the quiz grade a possible academic conduct case.

Your submission should not contain any main function. `main.cpp` is provided to you for testing.

Problems

1. String devil arc

Everyone is afraid of C-style strings, that is why string devil is so strong. But only way you can defeat a devil is by making people fear it less. You and your colleagues in public safety decided to practice with C-style strings and pointers enough so the string devil and also the pointers devil loses their power so chainsaw man can defeat them.

Your task is write a void function called *reverse* that has two parameters, a `char` pointer `str` and a `int n` that denotes the size of the string `str`. The function should in-place reverse `str`. That is you cannot create new array (dynamic or static) in your function nor return a new array, you have to edit and reverse the string directly.

Once going through this exercise you you will no longer fear C-style strings and pointers and should finally be able to defeat the string and pointer devil

2. Return of Sajid Lukman

After fighting for months against HR department of Sheep corporation, Sajid Lukman finally was able to get his full paycheck. But now he has mae some financial decisions that will eventually lead him to declare bankruptcy in the near future. Him being a huge weeb who has never touched grass in his life has spent all his money on mangas, comics and scifi and fantasy novels. To the point Sajid now has a sizeable collection. His financial adviser Rana Sahab suggests him to sell his collection so he can earn back some money that he can later spend on gambling. He decided to write a program that stores his collection, and calculate its worth.

Your task is to first write a `class` called `Book` that has the following private attributes:

- `string Name`
- `string Author`
- `int Volume`
- `int PublicationYear`
- `double Price`

The class should also have the following methods: `string N`, `string A`, `int V`, `int Y`, `double P`

- A default constructor
- A parameterized constructor that takes parameters for name, authors name, volume, year and publication year (in that order) and assign them to the respective attributes.

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- A method called *displayBook()* that returns a formatted string containing the book information. So if name = “Chainsaw man”, author = “Tatski Fujimoto”, volume = 1 and publication year = 2019 then the function should return “Chainsaw man vol. 1 by Tatski Fujimoto, 2019”.

You can add any getter or setter to help in your implementation.

next you hve to write a `class` called `BookShelf` that has the following private attributes:

- An `int` denoting how much space you have in your bookshelf.
- An `int` denoting how many books you currently have your bookshelf.
- A dynamically allocated array of Books that hold the books currently on your bookshelf.

The class should also have the following methods:

- A default constructor
- A parameterized constructor that takes a parameter for the space and create a bookshelf with that much space.
- A method called *AddBook()* that takes a parameter book `Book`, there is space on the book shelf, it should add that book to the shelf in the last available position. And then return a string confirming the book has been added so if a book with name = “Chainsaw man”, author = “Tatski Fujimoto”, volume = 1 and publication year = 2019 then the function should return “The Hobbit vol. 1 by J. R. R. Tolkien, 1937 added to the book shelf\n”. If there is not space on the shelf the function should return “There is no more space on the book shelf!\n”
- A method *ShowCollection()* that returns a string that is a concatenation of *displayBook()* of each book in the bookshelf, separated by “\n”. More formally you have to return x where

$$x = \sum_{i=0}^{\text{number of books}} books[i].displayBook() + “\n”$$

Here summation denotes string concatenation.

- A method calls *worth()* that computes and returns the total value of the collection.

You can add any getter or setter or other methods help in your implementation.

Disclaimer

This quiz is entirely a work of fiction. The names, characters and incidents portrayed in it are the work of the RA’s imagination. Any resemblance to actual persons, living or dead, events or localities is entirely coincidental